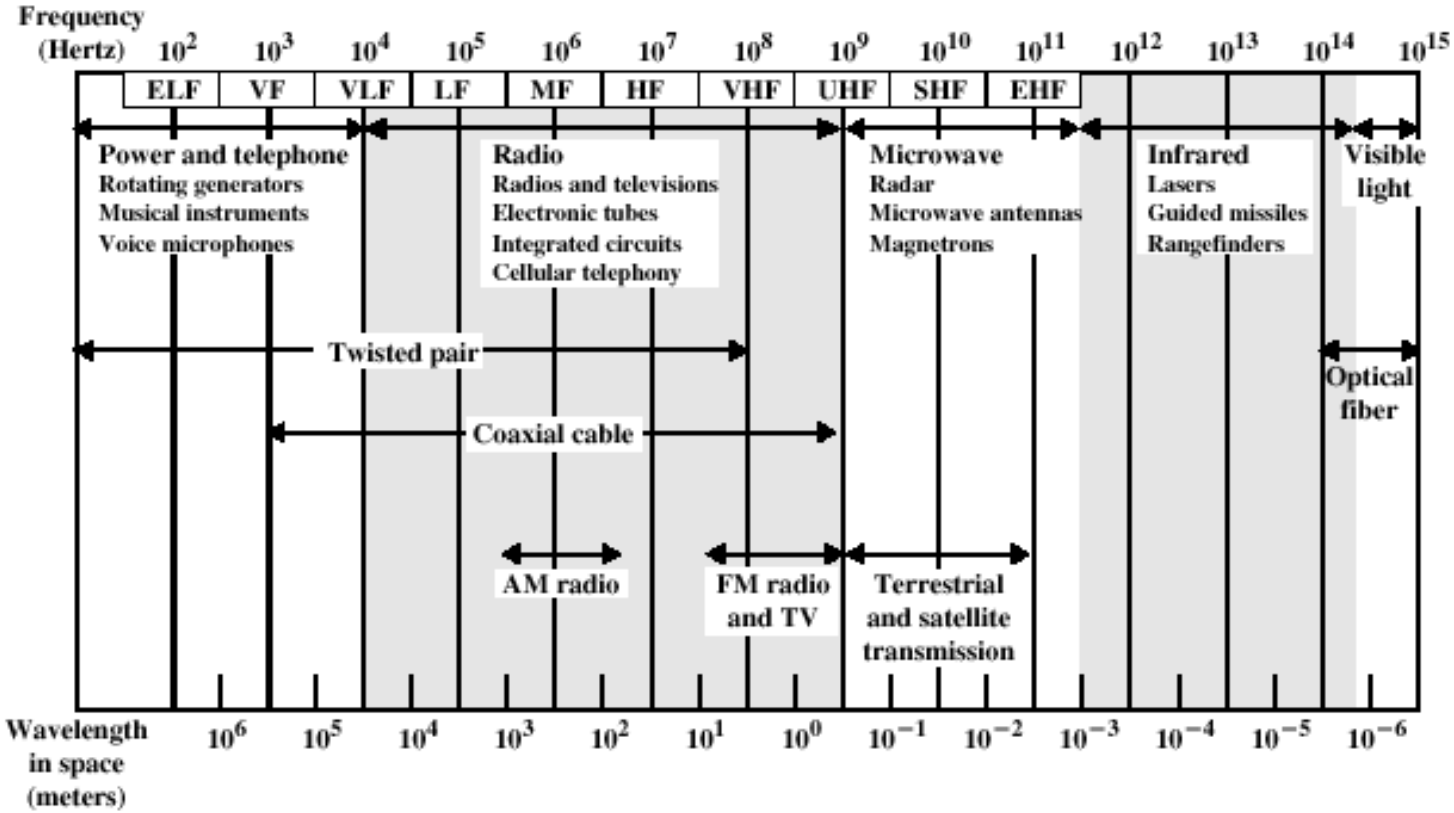
Chapter 16 Transmission Media

Transmission Media

Transmission medium: the physical path between transmitter and receiver.

- *Repeaters or amplifiers may be used to extend the length of the medium.*

Electromagnetic Spectrum for Transmission Media



Guided Media and Unguided Media

Communication of electromagnetic waves is guided or unguided.

Guided media:

•Waves are guided along a physical path

•Use a conductor such as a wire or a fiber optic cable to move the signal from sender to receiver

•Transmission capacity depends on the distance and on whether the medium is point-to-point or multipoint

Unguided media:

•Use radio waves of different frequencies and do not need a wire or cable conductor to transmit signals (e.g., the atmosphere and outer space).

Design Factors for transmission media

***Bandwidth**: All other factors remaining constant, the greater the band-width of a signal, the higher the data rate that can be achieved.*

***Transmission impairments:** Limit the distance a signal can travel.*

***Interference**: Competing signals in overlapping frequency bands can distort or wipe out a signal.*

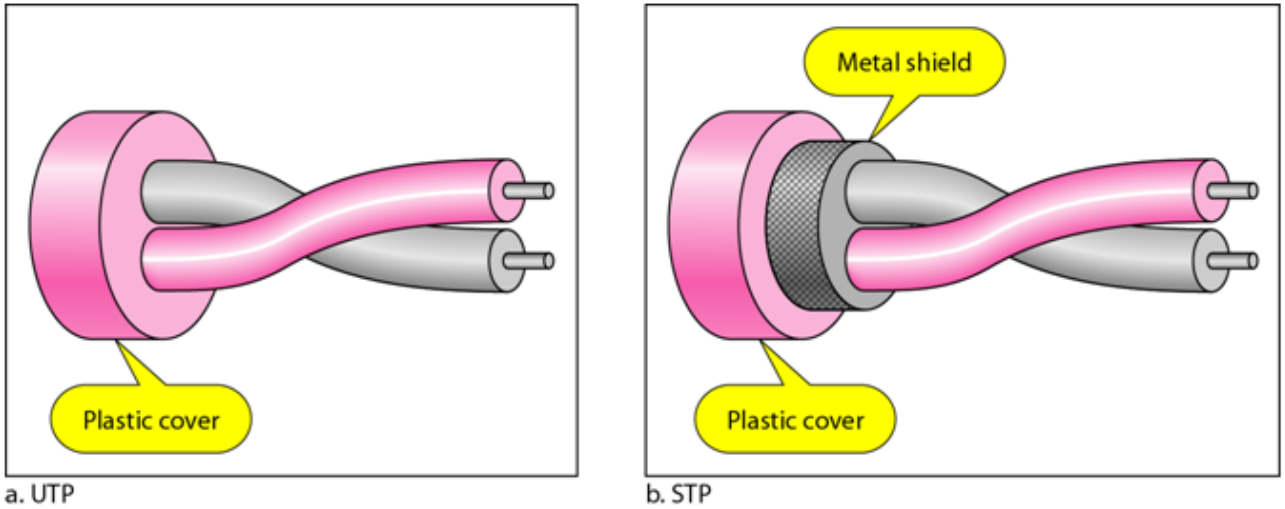
***Number of receivers**: Each attachment introduces some attenuation and distortion, limiting distance and/or data rate.*

Guided Transmission Media-Twisted Pair

- *Twisted Pair consists of two insulated copper wires arranged in a regular spiral pattern to minimize the electromagnetic interference between adjacent pairs.*
- *The signal is transmitted through one wire and a ground reference is transmitted in the other wire.*
- *Usually installed in building during construction.*

Types of twisted pair

- *STP (shielded twisted pair)*
  the pair is wrapped with metallic foil to insulate the pair from electromagnetic interference
- *UTP (unshielded twisted pair)*
  each wire is insulated with plastic wrap, but the pair is encased in an outer covering



a. UTP b. STP

Advantages and Disadvantages

- ***Twisted Pair Advantages***

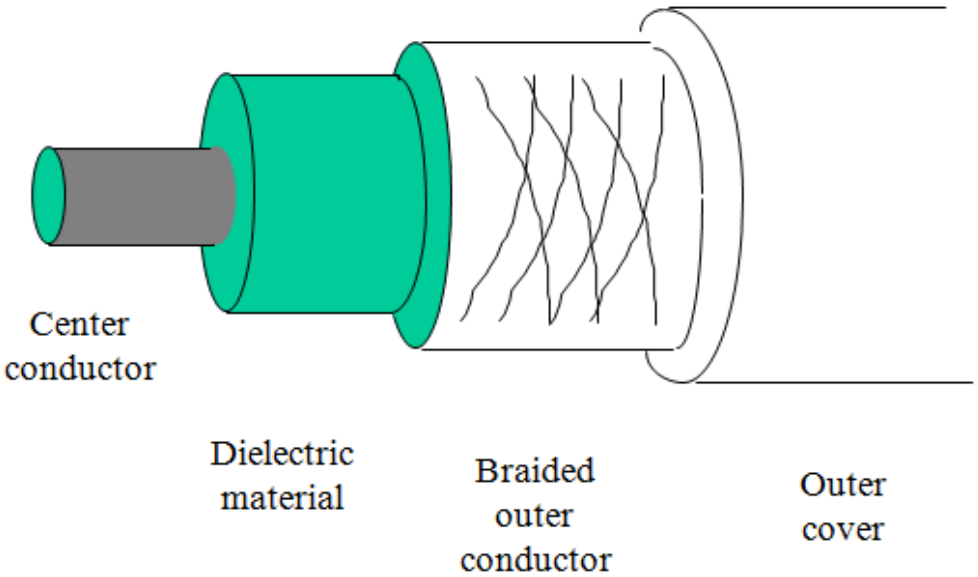
Inexpensive and readily available
Flexible and light weight
Easy to work with and install

- ***Twisted Pair Disadvantages***

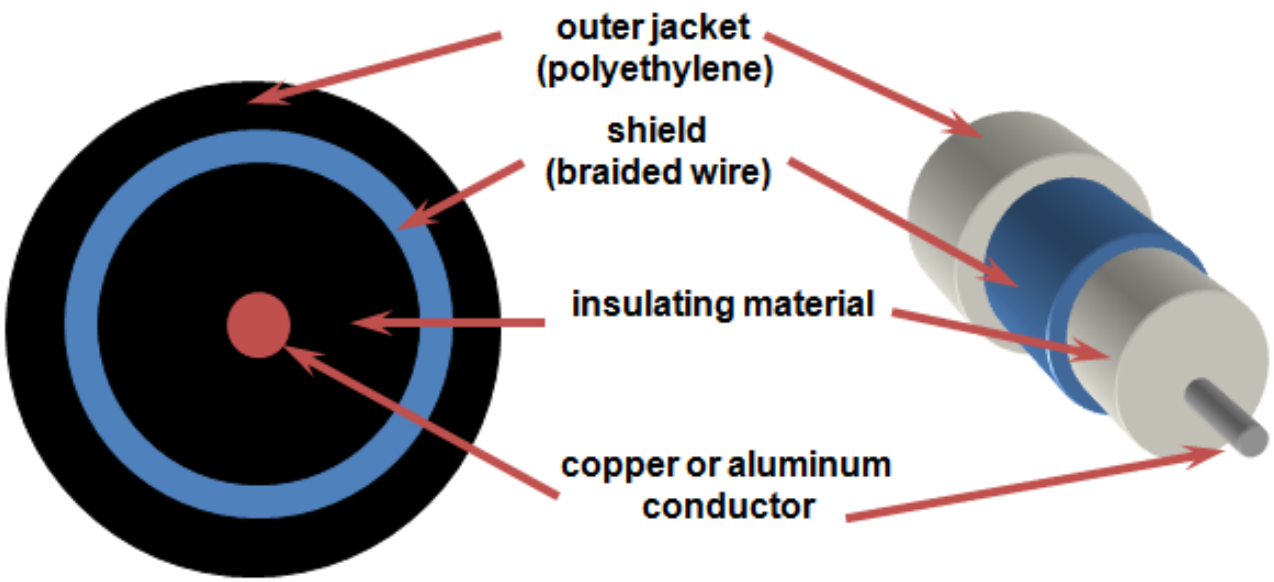
Susceptibility to interference and noise
Attenuation problem
    For analog, repeaters needed every 5-6km
    For digital, repeaters needed every 2-3km

Coaxial Cable

- *Used for cable television, LANs, telephony*
- *Has an inner conductor surrounded by a braided mesh*
- *Both conductors share a common center axial, hence the term "co-axial"*



Coaxial Cable



Advantages

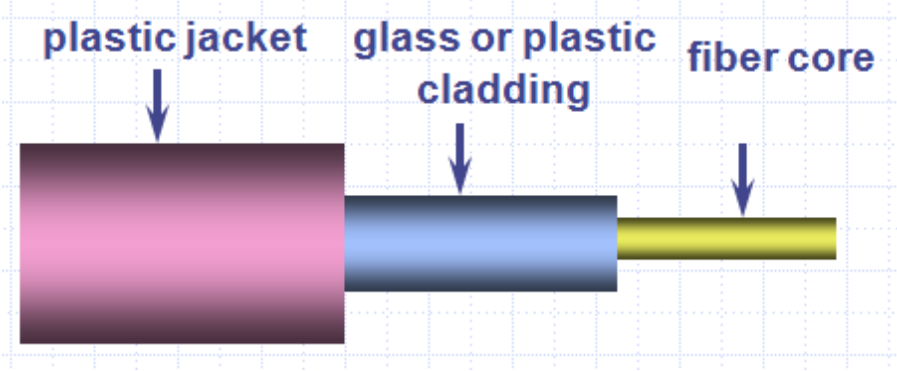
- *Higher bandwidth*
  - *400 to 600Mhz*
  - *up to 10,800 voice conversations*
- *Can be tapped easily (pros and cons)*
- *Much less susceptible to interference than twisted pair*

Disadvantages

- *More expensive to install compare to twisted pair cable.*
- *The thicker the cable, the more difficult to work with.*

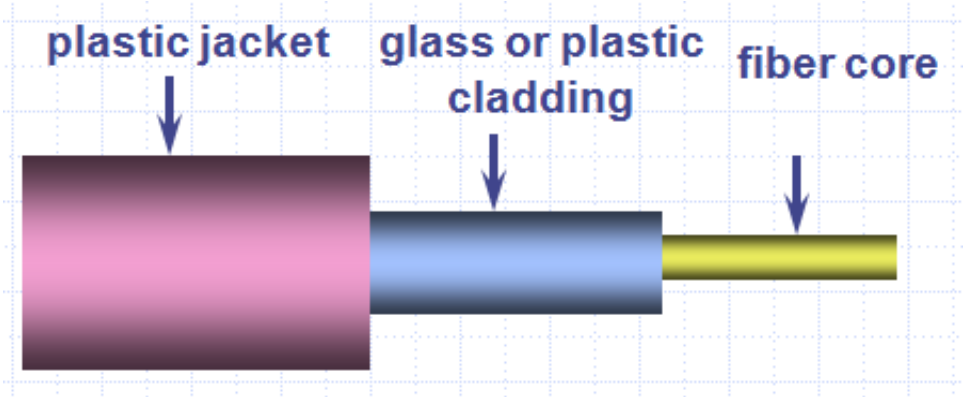
Fiber Optical Cable

Optical fiber is a thin flexible medium capable of conducting optical rays. Optical fiber consists of a very fine cylinder of glass (core) surrounded by concentric layers of glass (cladding).



- *Relatively new transmission medium used by telephone companies in place of long-distance trunk lines*
- *Also used by private companies in implementing local data communications networks*
- *Require a light source with injection laser diode (ILD) or light-emitting diodes (LED)*

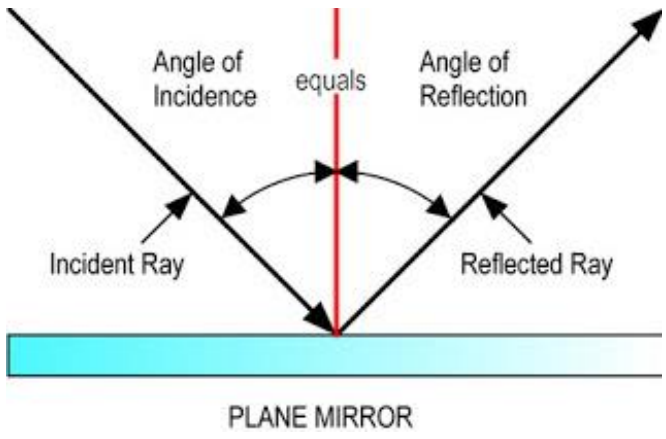
Fiber Optical Cable

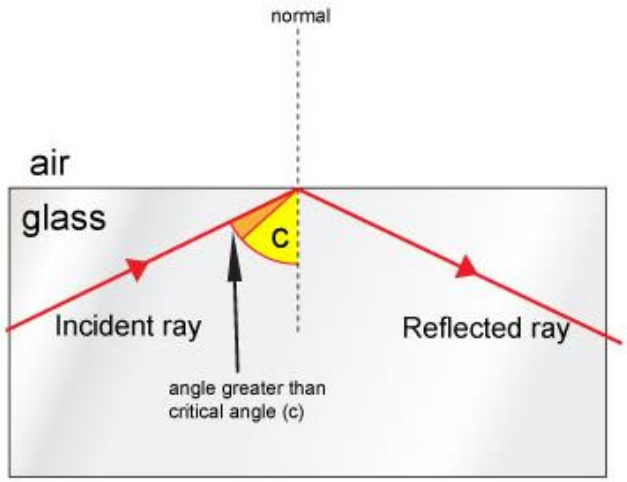


Optical fibers use reflection to guide light through a channel. A glass or plastic core surrounded by a cladding of less dense glass or plastic. The difference in density of the two materials must be such that a beam of light moving through the core is reflected off the cladding instead of being refracted into it. Information is encoded onto a beam of light as a series of on-off flashes that represent 1 and 0.

Fiber Optical Cable

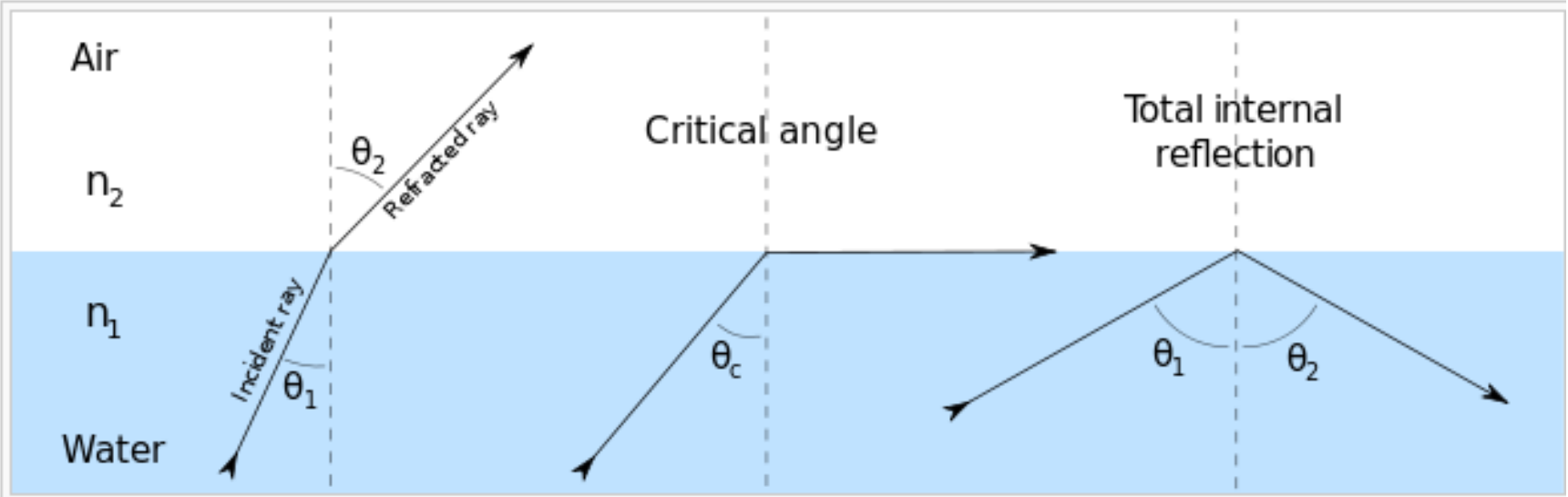
A signal-encoded beam of light (a fluctuating beam) is transmitted by total internal reflection. (The complete reflection of a light ray reaching an interface with a less dense medium when the angle of incidence exceeds the critical angle.)





Fiber Optical Cable

- *Total internal reflection occurs in the core because it has a higher optical density (index of refraction) than the cladding.*
- *Attenuation in the fiber can be kept low by controlling the impurities in the glass.*



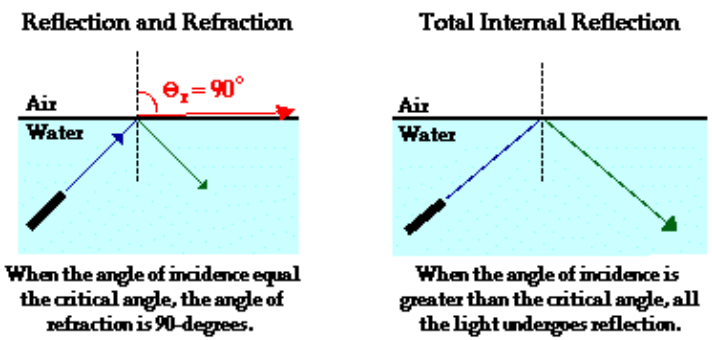
Refraction of light at the interface between two media, including total internal reflection.

What is Critical Angle?

•Total internal reflection (TIR) is the phenomenon that involves the reflection of the entire incident light off the boundary. TIR only takes place when both of the following two conditions are met:

•A light ray is in the denser medium and approaching the less dense medium.

•The angle of incidence for the light ray is greater than the so-called critical angle.

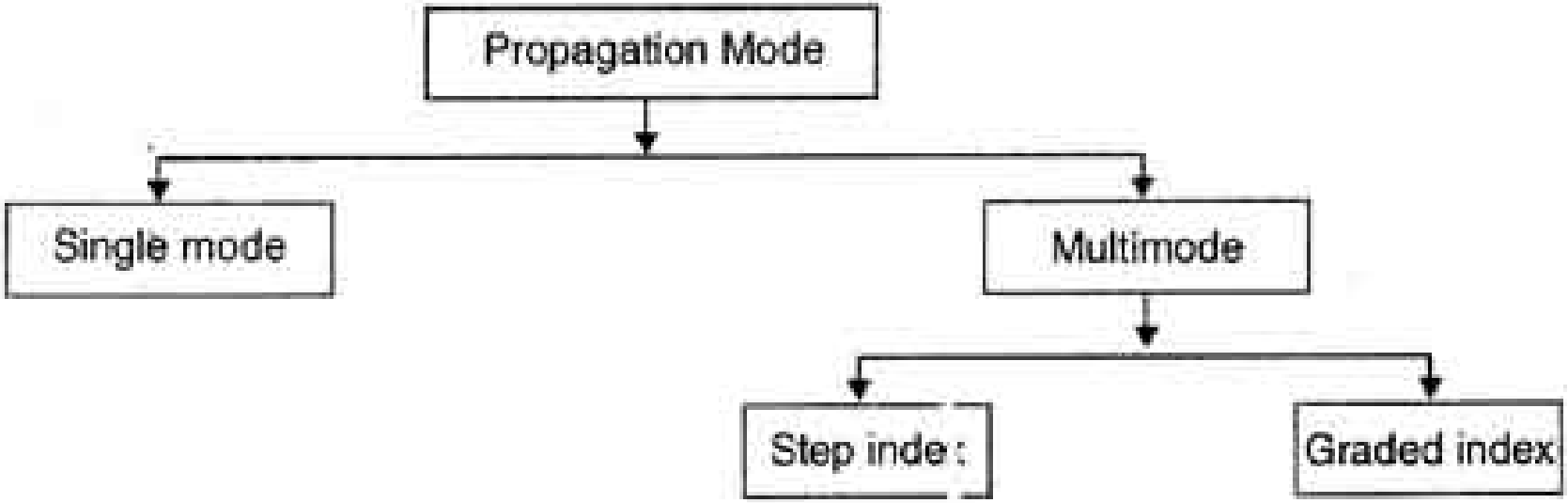


•Let us consider the example of light travelling through water towards the boundary with a less dense material such as air. When the angle of incidence in water reaches a certain critical value, the refracted ray lies along the boundary, having an angle of refraction of 90-degrees. This angle of incidence is known as the critical angle; it is the largest angle of incidence for which refraction can still occur.

What is Critical Angle?

So the critical angle is defined as the angle of incidence that provides an angle of refraction of 90-degrees. Make particular note that the critical angle is an angle of incidence value. For the water-air boundary, the critical angle is 48.6-degrees. For the crown glass-water boundary, the critical angle is 61.0-degrees. The actual value of the critical angle is dependent upon the combination of materials present on each side of the boundary.

Propagation Modes

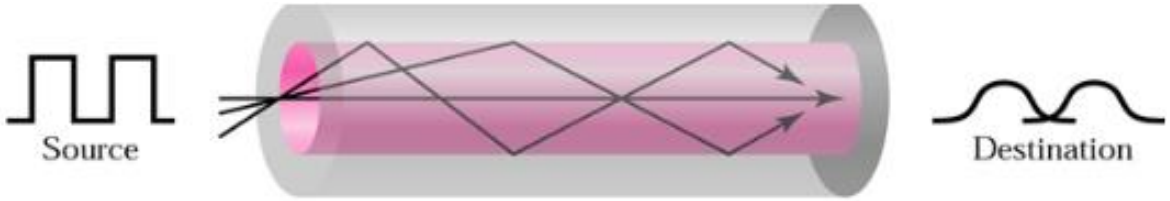


Multimode

Multimode is so named because multiple beams from a light source move through the core in different paths. How these beams move within the cable depends on the structure of the core.

Multimode step index

- *In multimode step-index fiber, the density of the core remains constant from the centre to the edges.*
- *A beam of light moves through this constant density in a straight line until it reaches the interface of the core and the cladding.*
- *At the interface, there is an abrupt change to a lower density that alters the angle of the beam's motion.*
- *The term step-index refers to the suddenness of this change.*



a. Multimode step-index

Multimode step index

a. Multimode, step-index

- *Some beams in the middle travel in straight lines through the core and reach the destination without reflecting and refracting.*
- *Some beams strike the interface of the core and cladding at an angle smaller than the critical angle; these beams penetrate the cladding and are lost.*
- *Still other hit the edge of the core at angles greater than the critical angle and reflect back into the core and off the other side, bouncing back and forth down the channel until they reach the destination.*

Multimode graded index

- *Multimode graded-index fiber, decreases the distortion of the signal through the cable.*

- *A graded-index fiber is one with varying densities.*

- *Density is highest at the centre of the core and decreases gradually to its lowest at the edge.*
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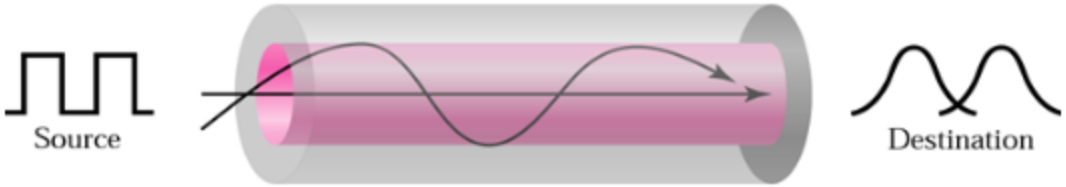
- *The impact of this variable density on the propagation of light beams is shown below:*

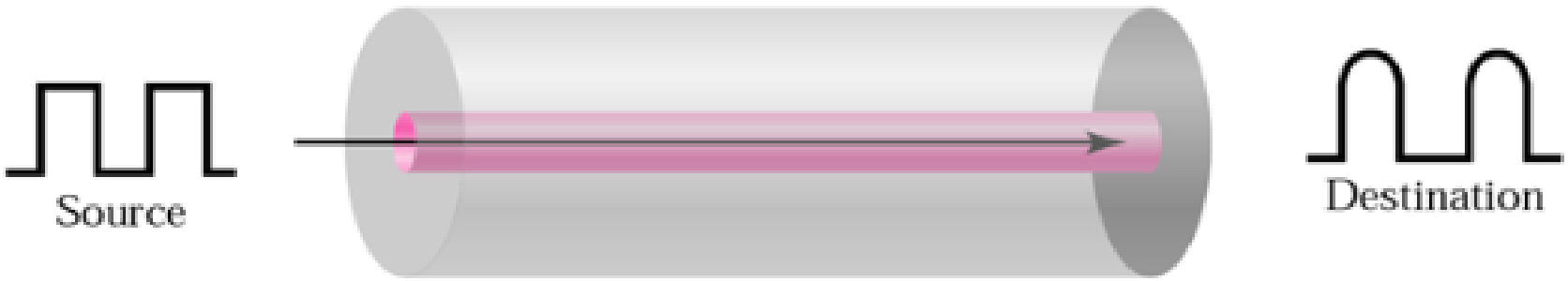


b. Multimode, graded-index

Single Mode

- *Single mode uses step-index fiber and a highly focused source of light that limits beams to a small range of angles and close to the horizontal.*
- *The single mode fiber itself is manufactured with a smaller diameter than that of multimode fibers, with substantially lower density.*
- *The decrease of density results in a critical angle that is close enough to 90 degrees to make the propagation of beams almost horizontal.*
- *In this case propagation of different beams is almost identical and delays are negligible.*
- *All of the beams arrive at the destination "together" and can be recombined without distortion to the signal.*

Single Mode



c. Single-mode

Advantages

- *greater capacity (bandwidth of up to 2 Gbps)*
- *smaller size and lighter weight*
- *lower attenuation*
- *immunity to environmental interference*
- *highly secure due to tap difficulty and lack of signal radiation*

Disadvantages

- *expensive over short distance*
- *requires highly skilled installers*
- *adding additional nodes is difficult*

Wireless Transmission

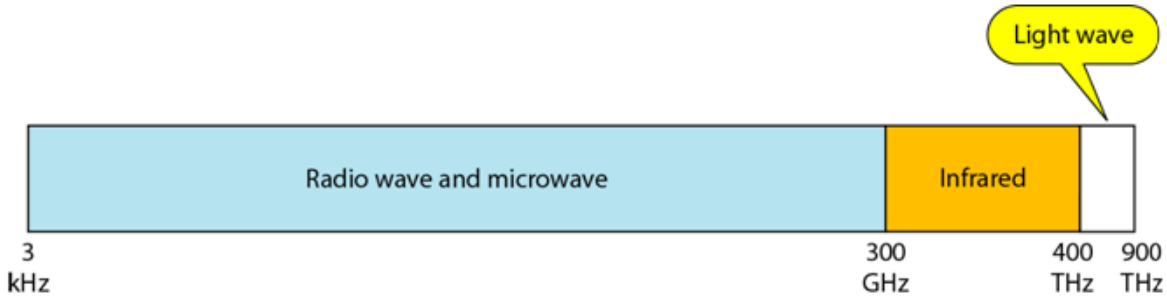
Transmission and reception are achieved by means of an antenna

- *directional*
  - *transmitting antenna puts out focused beam*
  - *transmitter and receiver must be aligned*
- *omnidirectional*
  - *signal spreads out in all directions*
- *can be received by many antenna*

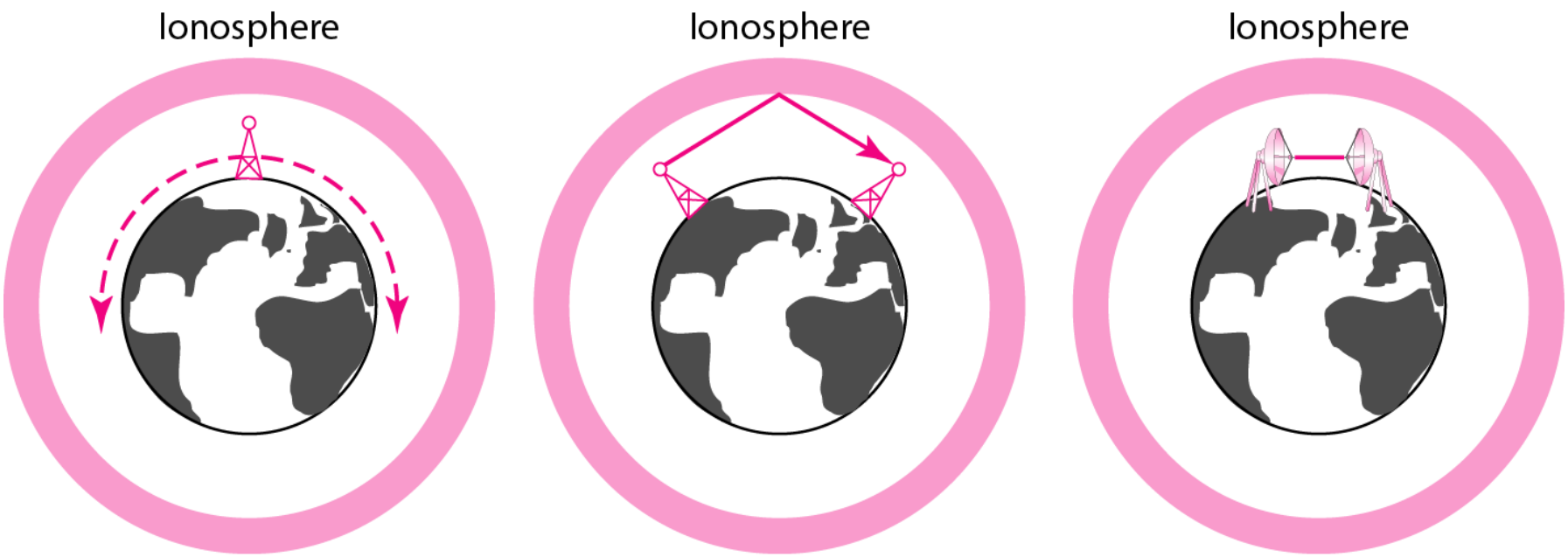
Wireless Example

- *Radio Wave*
- *Microwave*
- *Infrared*

Electromagnetic spectrum for wireless communication



Propagation method



Ground propagation
(below 2 MHz)

Sky propagation
(2–30 MHz)

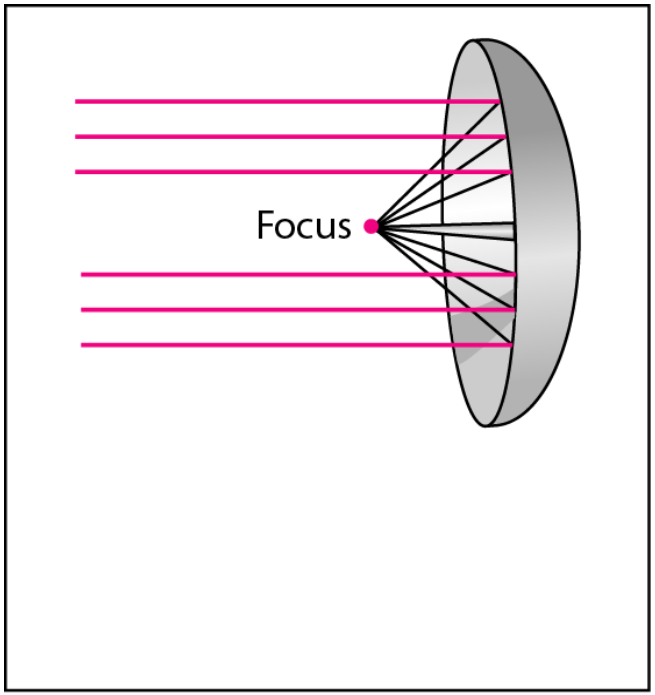
Line-of-sight propagation
(above 30 MHz)

Radio Wave

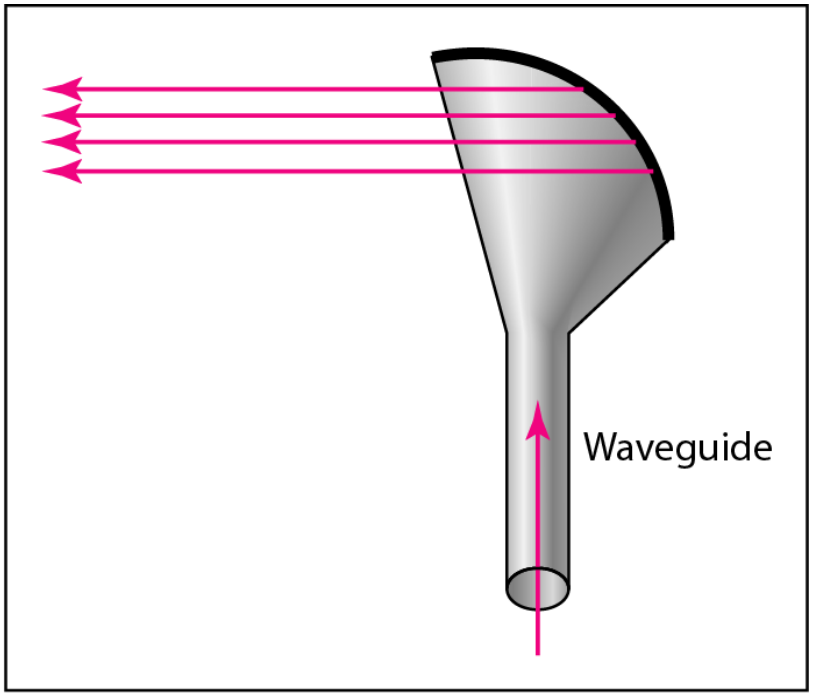
- *Used for multicast communications, such as radio and television.*
- *Can penetrate through walls.*
- *Highly regulated.*
- *Use omnidirectional antennas*

Microwave

- *Microwaves are used for unicast communication such as cellular telephones, satellite networks and wireless LANs.*
- *Higher frequency ranges cannot penetrate walls.*
- *Use directional antennas - point to point line of sight communications.*



a. Dish antenna



b. Horn antenna

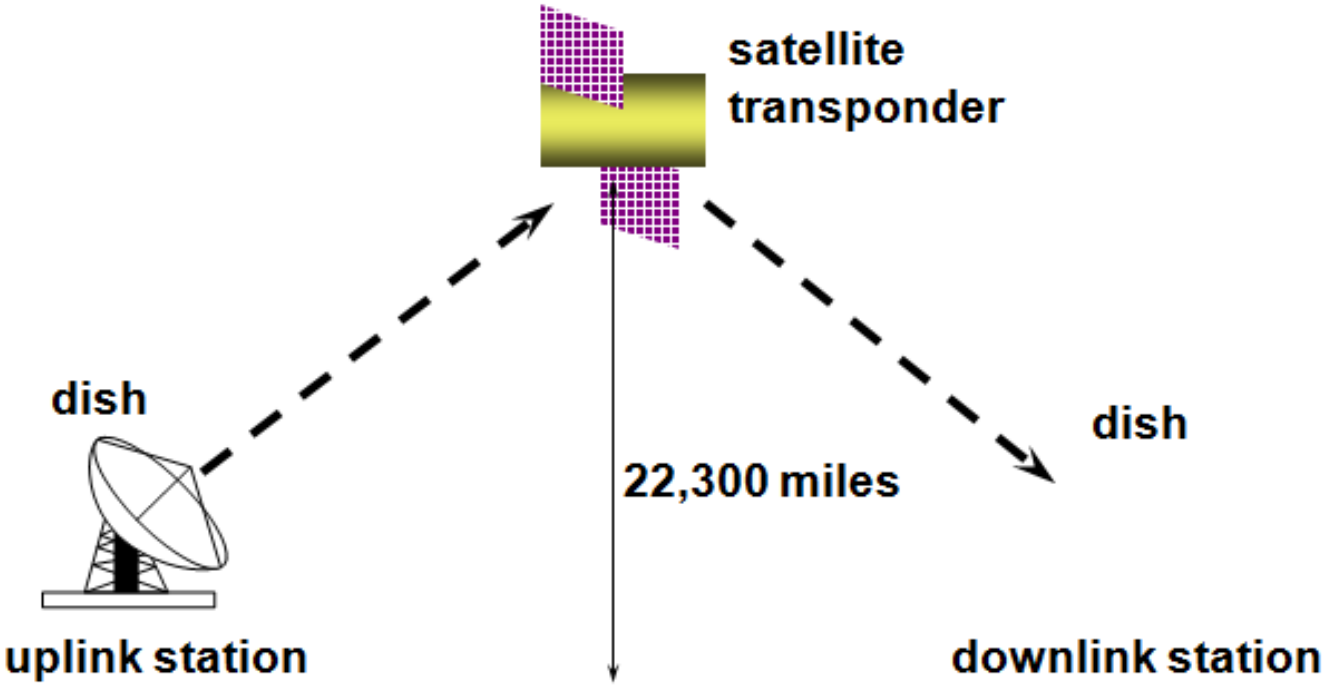
Terrestrial Microwave

- *used for long-distance telephone service*
- *uses radio frequency spectrum, from 2 to 40 Ghz*
- *parabolic dish transmitter, mounted high*
- *used by common carriers as well as private networks*
- *requires unobstructed line of sight between source and receiver*
- *curvature of the earth requires stations (repeaters) ~30 miles apart*

Satellite Microwave Transmission

- *a microwave relay station in space*
- *can relay signals over long distances*
- *geostationary satellites*
    - *remain above the equator at a height of 22,300 miles (geosynchronous orbit)*
    - *travel around the earth in exactly the time the earth takes to rotate*
- *earth stations communicate by sending signals to the satellite on an uplink*
- *the satellite then repeats those signals on a downlink*
- *the broadcast nature of the downlink makes it attractive for services such as the distribution of television programming*

Satellite Transmission Process



Principal Satellite Transmission Bands

- *C band: 4(downlink) - 6(uplink) GHz*
  - *the first to be designated*
- *Ku band: 12(downlink) -14(uplink) GHz*
  - *rain interference is the major problem*
- *Ka band: 19(downlink) - 29(uplink) GHz*
  - *equipment needed to use the band is still very expensive*

Satellite Microwave Application

- *Television distribution*
- *Long-distance telephone transmission*
- *Private business networks*

Satellite Microwave Disadvantages

- *line of sight requirement*
- *expensive towers and repeaters*
- *subject to interference such as passing airplanes and rain*

Infrared

- *Signals can be used for short-range communication in a closed area using line-of-sight propagation.*
- *Unlike microwaves, infrared does not penetrate walls*.